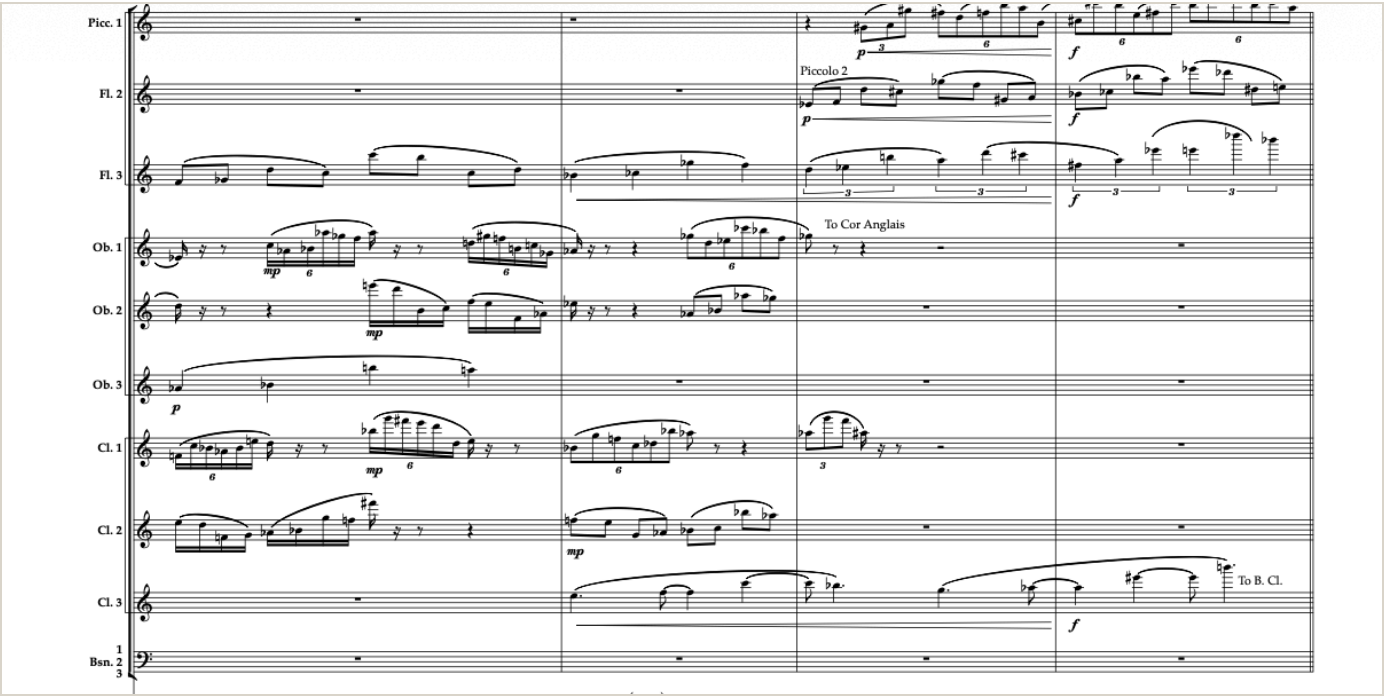FULL STACK PRODUCER (MUSIC NOTATION, AUDIO ML) · MUSE GROUP

Notation portfolio

Marc LeMay, PhD · Music Notation Specialist



I wrote, engraved, and proofed every page that is attached here for performance. Each one has a short note on what a recognition model would need to read correctly, which is also how I would review your models' output.

Below:

02 **Method.** The rubric I'd use for OMR and audio-to-score QA, and how it scales from a sprint check to a quarterly cycle.

03 **Sample evaluation.** Three of my pages run through MuseScore's PDF import, scored against the originals.

04–11 **Score pages.** Voice and piano · string quartet · orchestra · six voices with IPA · mixed ensemble with tenor · tuba and percussion · a mobile form

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02 · METHOD

Scoring rubric

I read every score against the same categories and three severity levels. This keeps a five-score sprint check and a 300-score quarterly cycle comparable from one release to the next.

SEVERITY

Critical	A musician would play or sing something wrong: pitch, accidental, duration, clef, key, meter, a tie read as a slur.
Major	The notes are right but something the player needs is missing: a dynamic, articulation, tuplet bracket, tempo text or lyric syllable.
Minor	Cosmetic, or obvious from context: stem direction, spacing, text placement. Itemized in a single-page sample like page 03. At sprint volume, counted only.

CATEGORIES

Pitch and accidental · Rhythm and duration · Tuplet · Tie and slur · Clef, key and meter · Barline, repeat and structure · Staff and voice assignment · Articulation and ornament · Dynamics and hairpins · Text, tempo and technique · Lyrics · Missing object · Phantom object · Layout

Each finding is logged as *measure · staff · category · severity · original shows · output shows*, so an engineer can find it in one look.

BY CYCLE

Sprint release 5–20 OMR, 5–10 A2S	Every Critical itemized, Majors itemized, Minors counted. A short written verdict per release: better than last time, worse, or the same, and where.
Quarterly A/B 100–300 scores	Both candidates scored blind on the same set. Results reported as differences by category and severity, so it is clear which categories improved and which got worse.
Benchmarking	Competitors scored on the golden set with the same rubric, so the comparison holds across products.
Ground truth	When the reference file the output is compared against turns out to be wrong, not the model, I flag it as a dataset issue so it doesn't count against the model.
Autograde	Each recording annotated against the score: what should have been played, what was played, the error type (wrong pitch, missed or extra note, early, late, wrong octave), and where. I taught this kind of error detection in aural skills courses for twelve years.

My pages through MuseScore's PDF import

Pages imported on October 11, 2026, and scored against my engraved originals using the rubric on page 02. Errors are grouped by root cause below, because one failure, such as a misread run of ledger lines, can produce many wrong notes.

Map You, p. 3 · voice and piano · 9 measures (mm. 14–22)

Critical 15 **Major 8** **Minor 10**

Verdict. Rhythm and the pitches inside the staff are strong. Grace-note groups and notes on high ledger lines are weak, accounting for 13 of the 15 Critical errors. Text is often misread.

Grace-note groups	**Critical**	All 6 groups in mm. 14–16 misread. One became a quarter rest; the others came out with wrong pitches (F♯–D–A read as G♯–D–B). In m. 16 the chord after the group lost two notes and its ties.
High ledger lines	**Critical**	Piano right hand, mm. 17–21: 6 wrong pitches or accidentals (E♭ read as F♯, D♮ as D♭, C♭ as C), plus a run of four tied notes read as three.
8va line	**Critical**	m. 17: the pitches under the line were transposed correctly but the line itself wasn't drawn, and the first two notes of beat 6 came out an octave high.
Text read as other words	**Critical**	Lyric "lines" read as "fores" (m. 16), so the singer would sing the wrong word. Also "pulsing" as "picking" and ♩ = 72 as 73 (Minor).
Not imported	**Major**	Rehearsal mark A, tempo text, a breath mark, a staccato, an accent and a slur. Dynamics moved below the vocal staff and below the piano left hand.
Cosmetic	**Minor**	Courtesy naturals dropped throughout, unison noteheads split, a short arpeggio sign, no title or measure numbers.

Process note. Map You took 40 minutes, half on paper and half typing into the log. For Entanglements AI transcribed my photographed notes and I checked every row by sight: 28 minutes end to end, about a third off the cost per score.

Entanglements, p. 10 · string quartet · 10 measures (mm. 85–94)

Critical 12 **Major 3** **Minor 4**

Verdict. The import didn't recognize a string quartet. It read the page as a grand staff, so whole measures of viola and cello went missing or had their music placed in the wrong clef. It also dropped every accidental written above a trill, the specificity of which is crucial in a non-tonal piece like this.

Instrumentation	**Critical**	Four instruments read as two staves, both labeled Vln. 1, with the wrong clefs. Viola mm. 88–90 and cello mm. 88–94 are missing entirely.
Trill accidentals	**Critical**	Every accidental above a trill sign was dropped, so the player trills to the wrong note. In m. 91 a tied, trilled B♭ 6 came out as a plain B♭ 5 with no trill and no tie.
Tremolo	**Critical**	Lost in both violins in m. 90 and in Vln. 2 in m. 91 (where D♮ was also read as D♭). A measured tremolo across two beats in mm. 92–93 came out as an unmeasured tremolo, then plain notes.
Rhythm and tuplets	**Critical**	m. 87: the Vln. 2 rhythm on beats 1–4 and the viola's quintuplet sixteenths were both misread. Also one wrong pitch (cello D♯ as D♮, m. 85).
Dynamics	**Critical**	Most markings on the page were dropped. Scored Critical rather than Major because a quartet with no dynamics can't shape the passage.
Not imported	**Major**	An accent and a tenuto (m. 87) and a courtesy natural (m. 93).
Cosmetic	**Minor**	Trill lines too short, rehearsal mark M misplaced, no title or measure numbers, and a down-bow added that isn't in the original (m. 94).

These Violent Delights, p. 23 · full orchestra · 28 staves
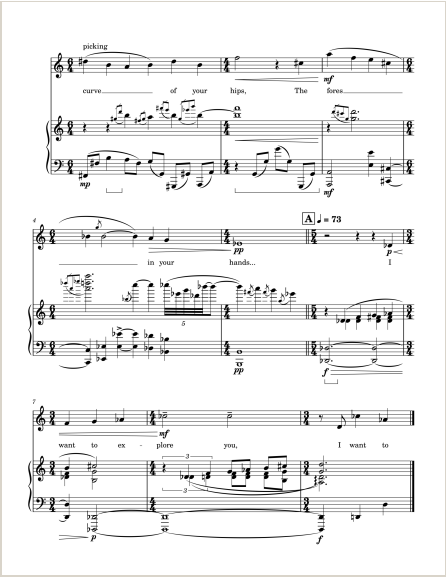
Stopped early

Verdict. The import didn't recognize which instrument each staff belongs to. Once that fails, every pitch and rhythm below it is unreliable, so I stopped itemizing and reported the root cause.
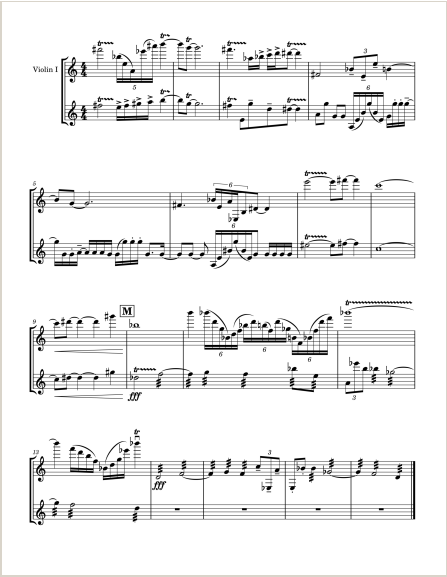
Part labels	**Critical**	Every staff was labeled with the wrong instrument. The last four staves had no label at all.
Missing staves	**Critical**	Two of the 28 staves were dropped entirely.
Notes and rhythm	**Not scored**	Widely wrong at a glance. Itemizing them would add nothing to the two findings above.

Triage note. On the voice and piano piece the import mostly misread notes. On the quartet and the orchestra pieces it failed earlier, at matching staves to instruments, so that's where I'd look first.
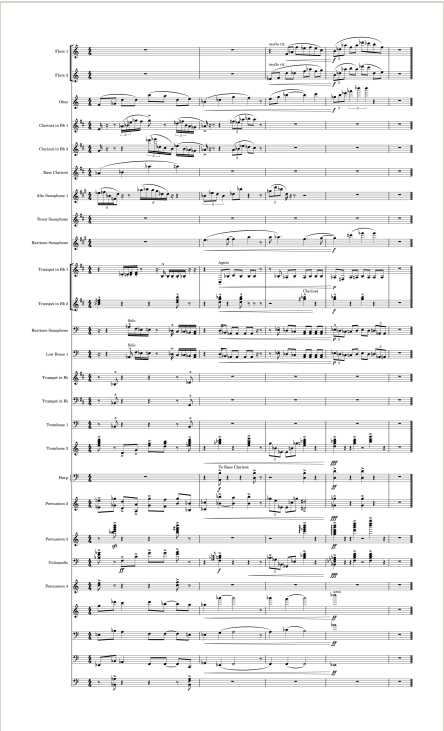
THE IMPORTS, AS MUSESCORE PRODUCED THEM
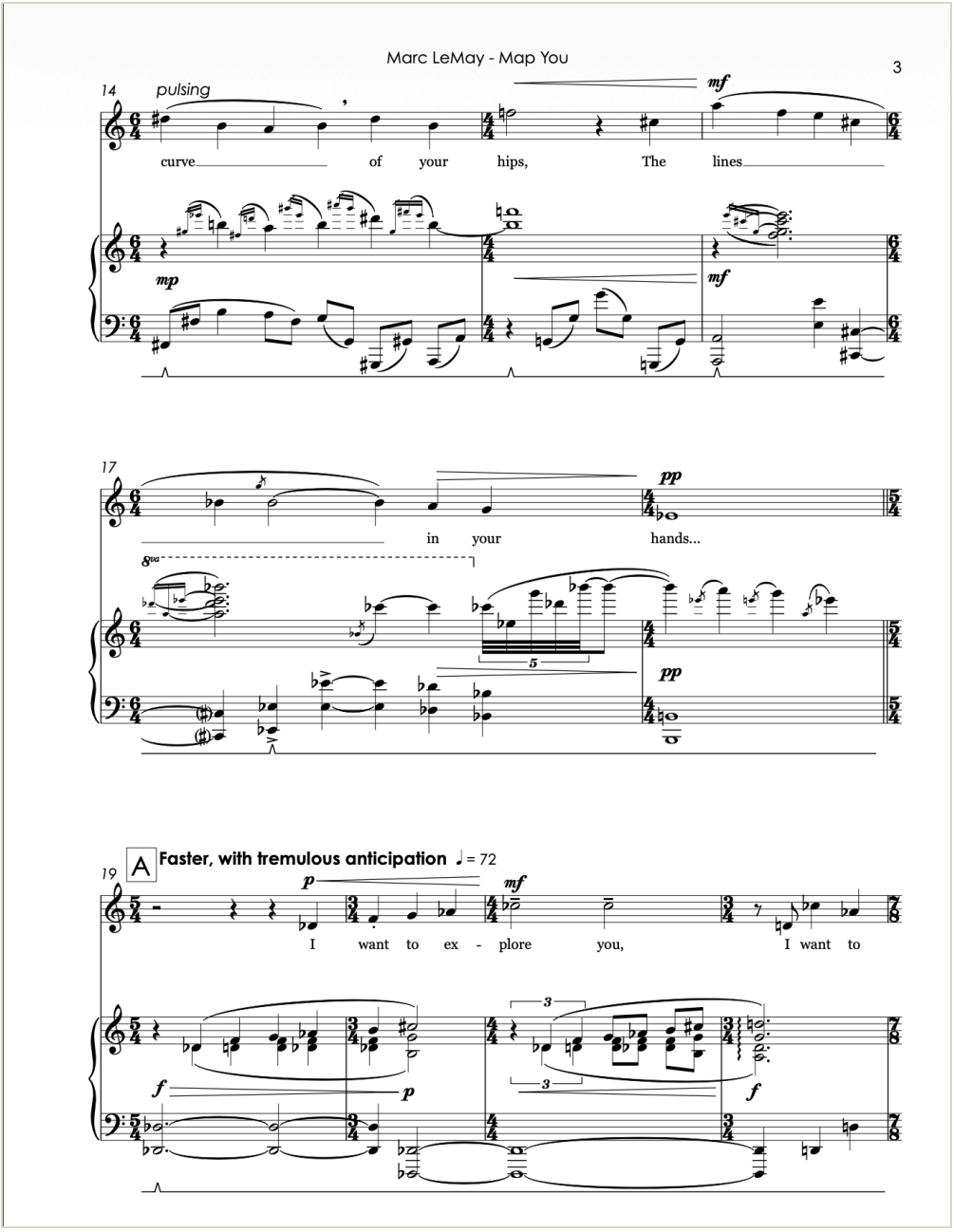


Map You, p. 3. Compare page 04.



Entanglements, p. 10. Four instruments on two staves. Compare page 05.



These Violent Delights, p. 23. Compare the cover.
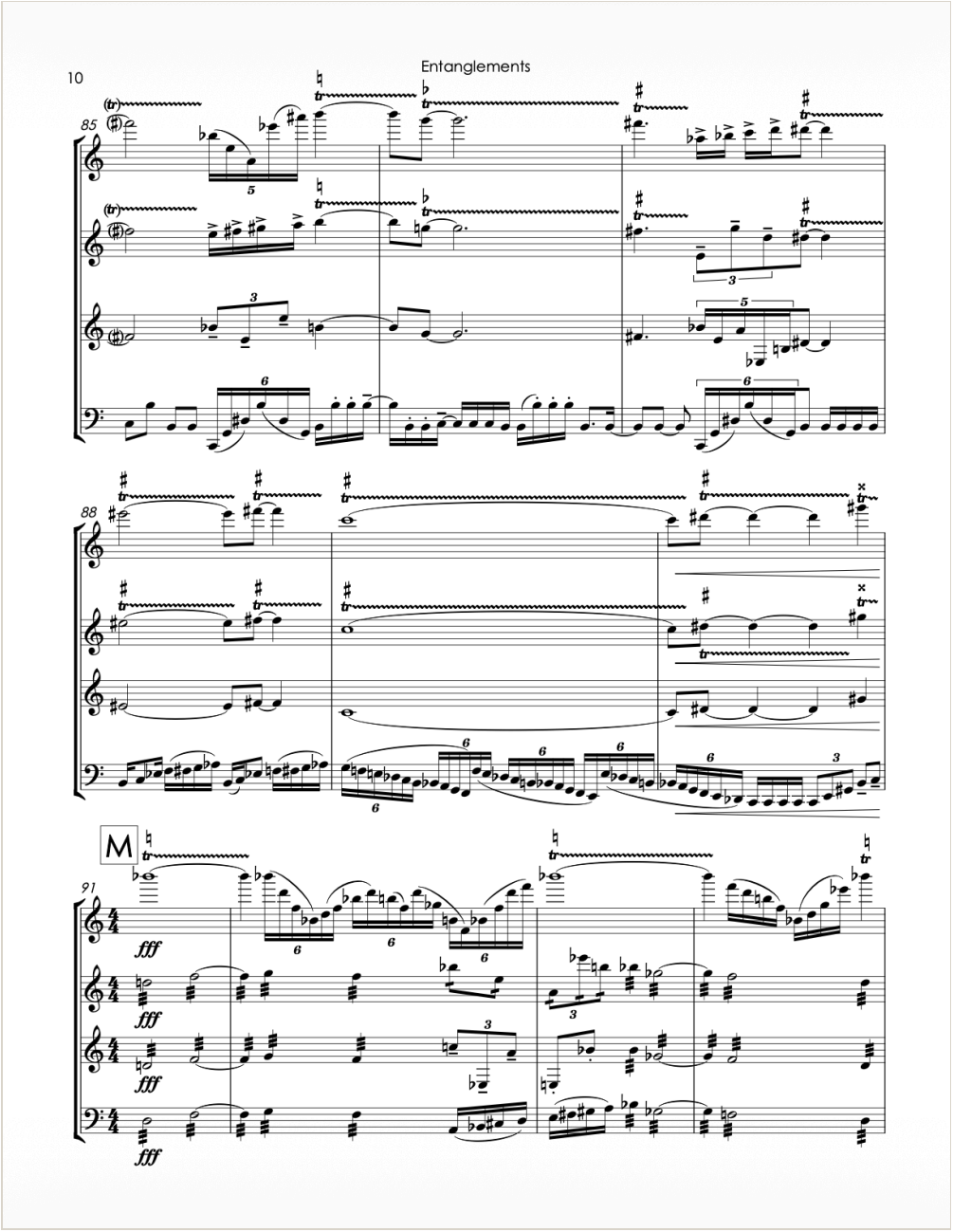


04 · VOICE AND PIANO

Map You, p. 3

mm. 14–22 · art song

WHAT A MODEL HAS TO GET RIGHT

- Lyrics with melismas and extender lines, attached to the right note.
- A new meter almost every bar.
- Grace notes before chords, and a quintuplet beamed to an 8th note.
- An 8va line over part of the right hand only.
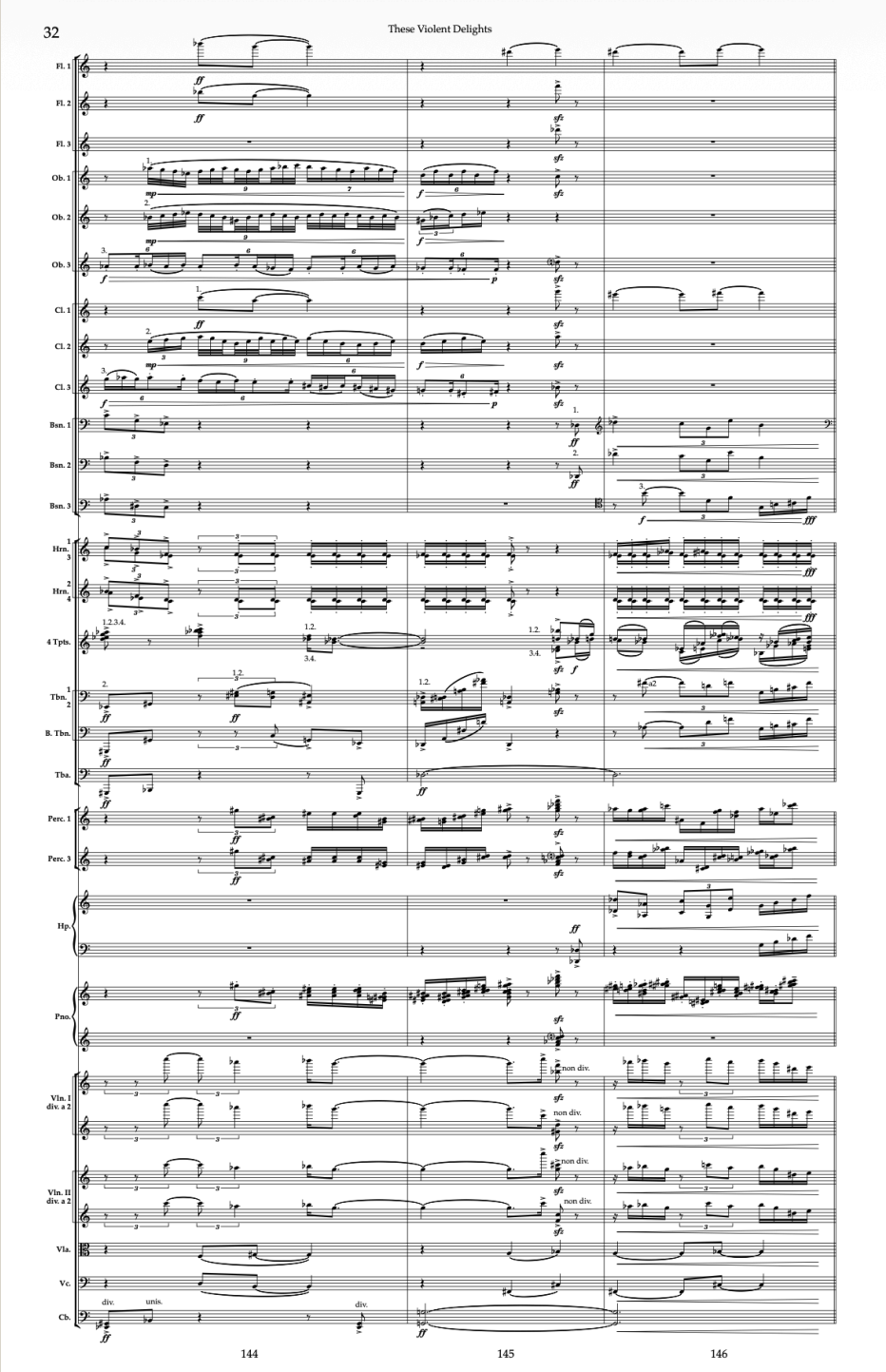


05 · STRING QUARTET

Entanglements, p. 10

mm. 85–94 · 2015 · premiered by the Aizuri Quartet

WHAT A MODEL HAS TO GET RIGHT

- Trills with an accidental above the sign, which applies to the trill note.
- Trill lines that cross barlines and end where the next trill begins.
- Quintuplets and sextuplets whose slurs cross the beam groups.
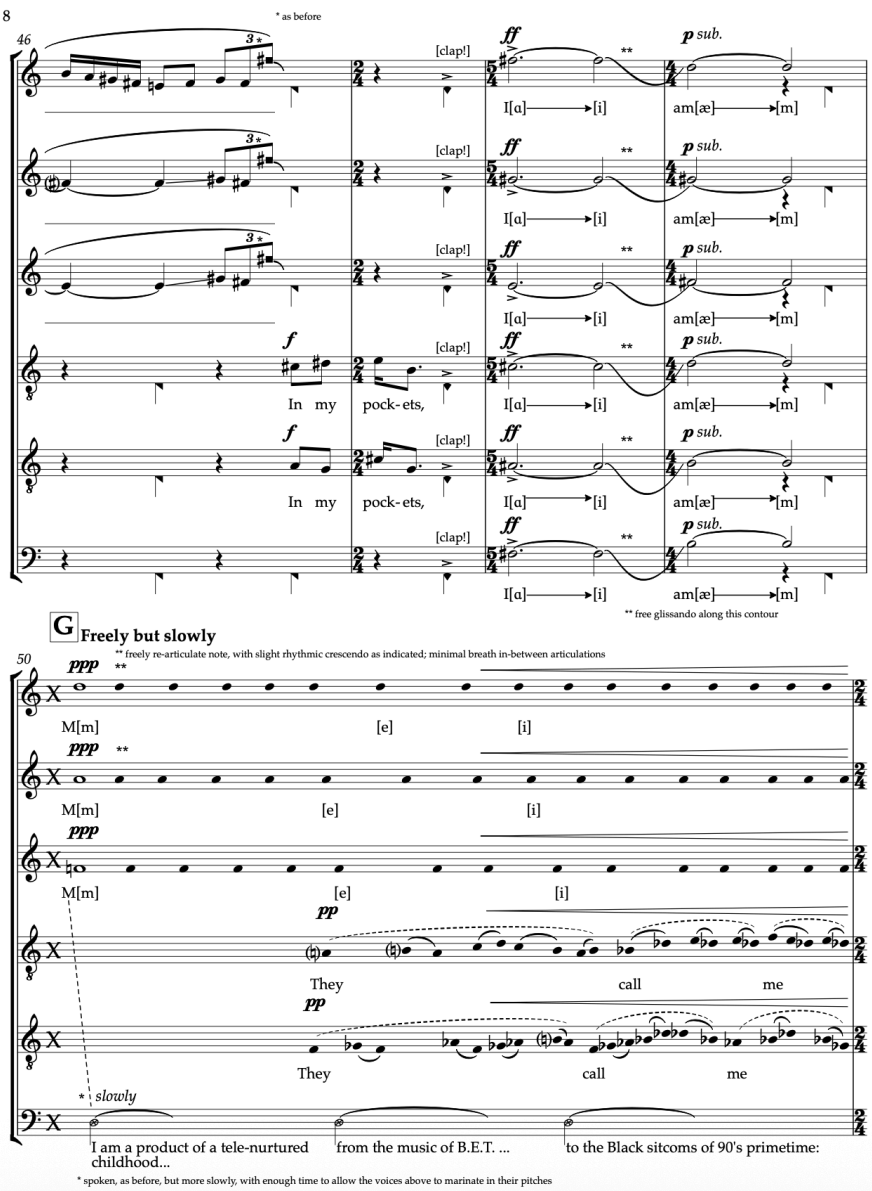


06 · ORCHESTRA

These Violent Delights, p. 32

mm. 144–146 · full orchestra · 2017 · read by the Curtis Symphony Orchestra

WHAT A MODEL HAS TO GET RIGHT

- Thirty-one staves on one page, with player numbers (1., 2., 3., a2) inside shared staves.
- Four trumpets on one staff, splitting into two voices and back.
- Divisi violins on paired staves, then “non div.” in the middle of a bar.
- Clef changes in the middle of a system: bassoon to treble, tenor clef in bassoon 3.
- Nine and seven against six and three in the woodwinds.



07 · SIX VOICES

INNERVISIONS of a Trebled Soul in a Troubled America, p. 8

mm. 46–50 · 2017 · premiered by Variant Six

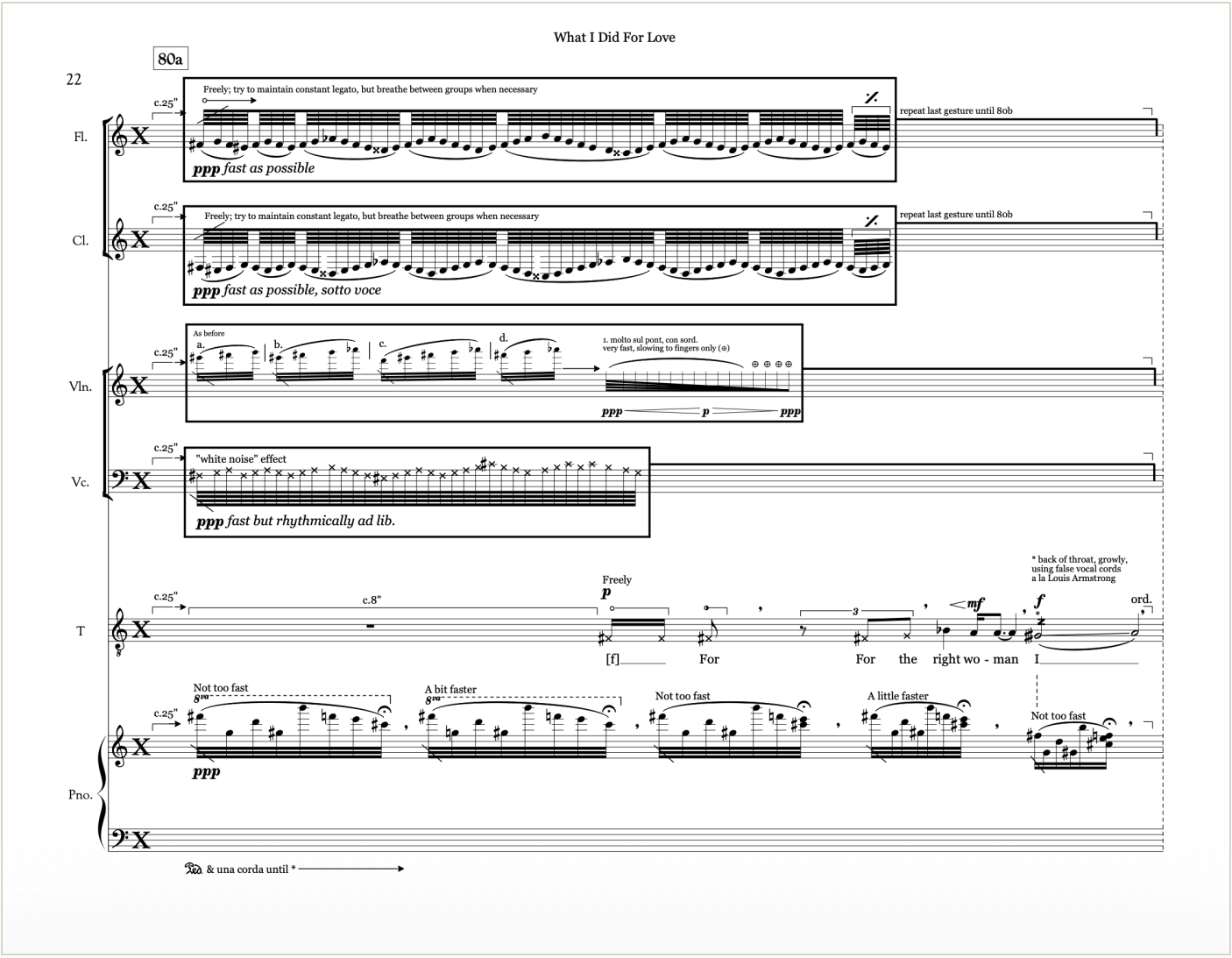
WHAT A MODEL HAS TO GET RIGHT

- IPA vowel changes in the lyric line, such as [ɑ] → [i].
- Glissando contours drawn between noteheads.
- Unmeasured time: an X where the meter would be, and stemless repeated notes.
- Spoken text under a single sustained x-notehead in the bass.

08 · TENOR AND MIXED ENSEMBLE

What I Did For Love, p. 22

Rehearsal 80a · 2018 · commissioned and premiered by the CHAI Collaborative Ensemble, Chicago · text by Darrel Holnes



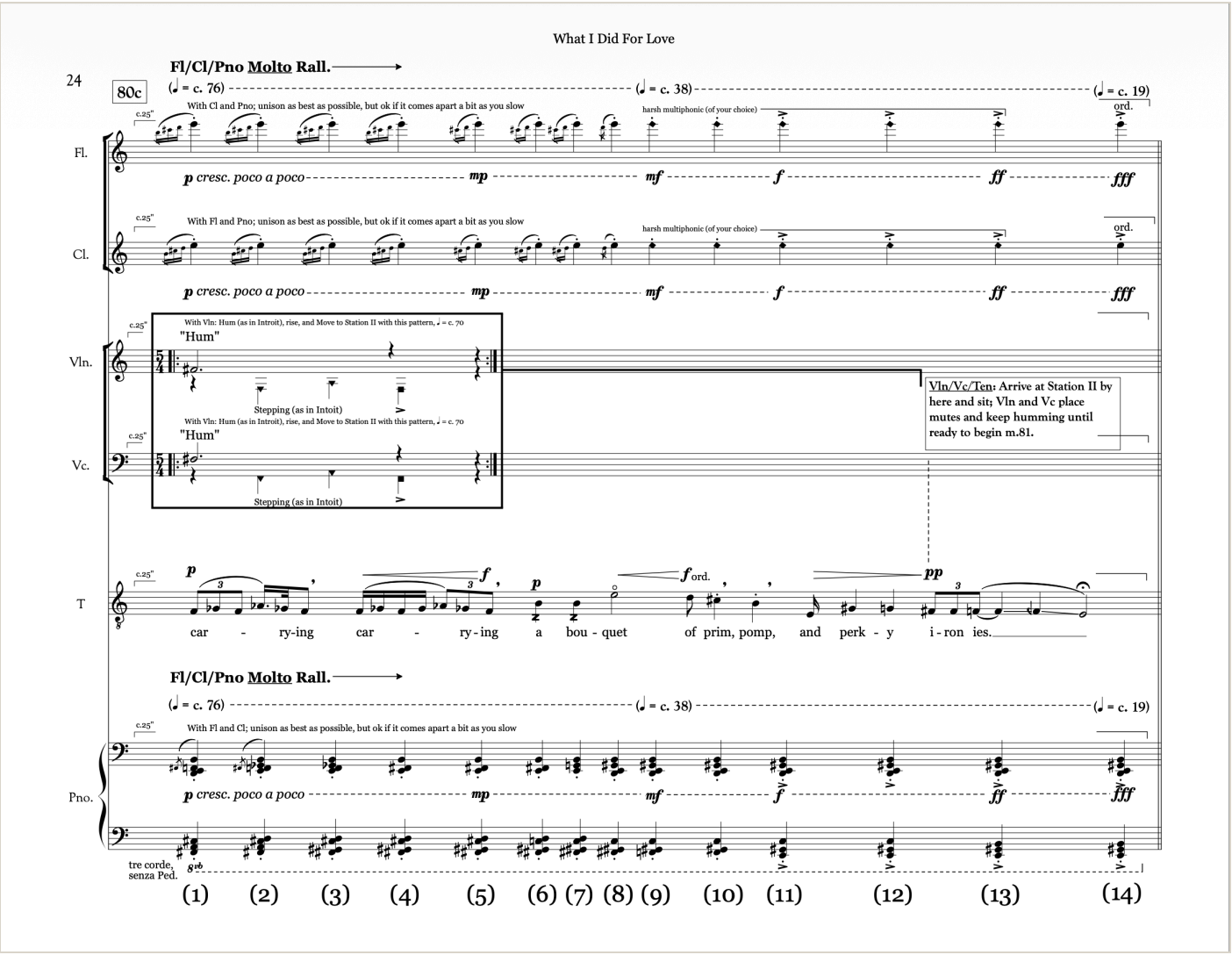
WHAT A MODEL HAS TO GET RIGHT

- No meter: timings in seconds, with boxed material played freely.
- x-noteheads mixed with pitched notes on the same staff.
- Feathered beams indicating an accelerando.
- A repeat-gesture sign and “repeat until 80b” in place of written bars.

09 · TENOR AND MIXED ENSEMBLE

What I Did For Love, p. 24

Rehearsal 80c · 2018
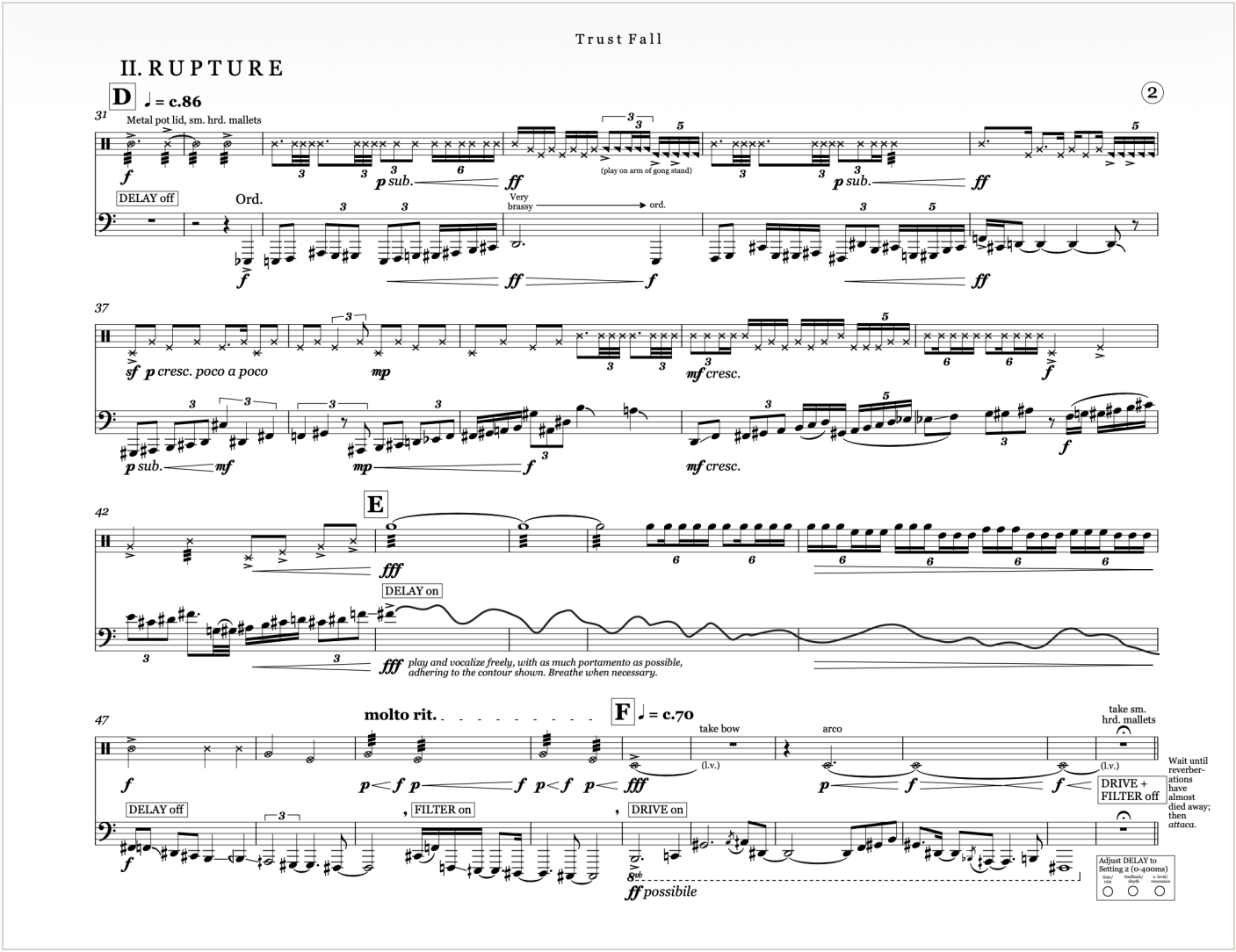


WHAT A MODEL HAS TO GET RIGHT

- Two simultaneous tempos: the trio slows while the strings loop a boxed 5/4 pattern.
- Boxed staging directions alongside technique text.
- Numbered pulse counts under the piano, which could be misread as fingering.
- The tenor's lyrics, breath marks and harmonics, inside a gradual tempo change.

10 · AMPLIFIED TUBA AND PERCUSSION

Trust Fall, II. Rupture

mm. 31–55 · 2019 · commissioned and premiered by BrassTaps Duo, New World Center, Miami



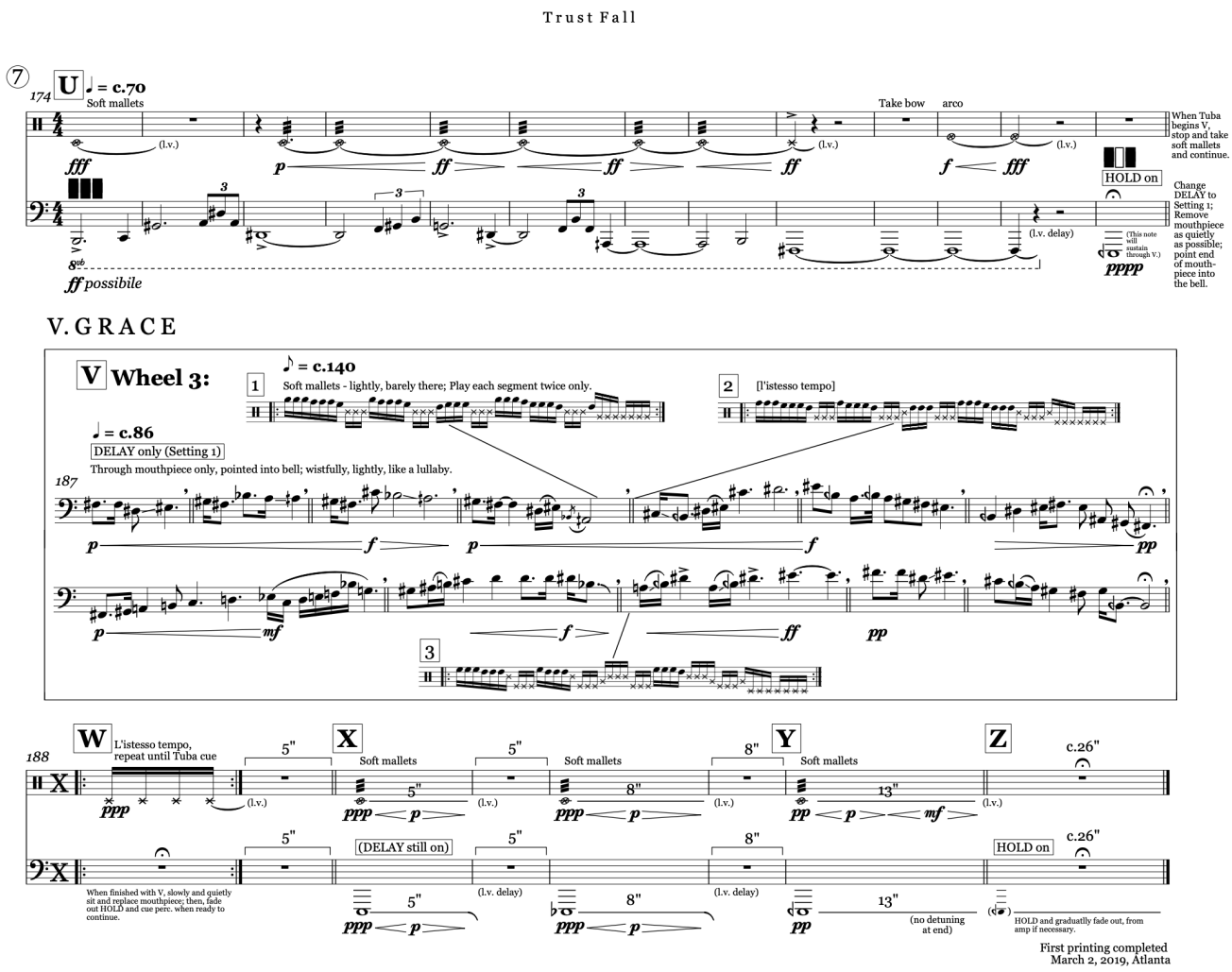
WHAT A MODEL HAS TO GET RIGHT

- Unpitched percussion with different noteheads (x, circled x, normal) for different instruments.
- Boxed effects-pedal cues (DELAY, FILTER, DRIVE) to be read as text.
- A drawn pitch contour in place of notes, for four bars.
- A long 8vb line affecting every pitch beneath it.

11 · AMPLIFIED TUBA AND PERCUSSION

Trust Fall, V. Grace

mm. 174–188 · 2019



WHAT A MODEL HAS TO GET RIGHT

- A mobile form: three looping percussion cells, each joined by a line to its entry point in the tuba part.
- Two tempo marks working at once inside the box.
- Bars measured in seconds (5″, 8″, 13″) instead of beats.
- Performance text (HOLD, DELAY, “through mouthpiece only”) that has to be kept as text.